

EFFECTIVE MODELING IN A CHANGING ENVIRONMENT WITH BAYESIAN MCMC METHODS. AN APPLICATION EXAMPLE FROM THE OIL INDUSTRY

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GOALS

1. Promote using Bayesian MCMC estimation methods
2. Introduce its application in SAS on a simple example
3. Compare Bayesian MCMC and Classical (Frequentist) estimation results

PLAN

1. Problem definition
2. Approaches to solving the problem
3. Classic vs. Bayesian estimation approaches
4. Application example

PROBLEM NEW DATA DO NOT FOLLOW OLD DATA MODELS

Examples:

1. New products (shale gas, new oil well, new flu strains);
2. Paradigm shifts caused by inventions (PC, Internet, E-commerce, shipping containers);
3. Government regulations (interest rates, amortization period);
4. Regime switches caused by new products (recession caused by USA CMO in 2008).

APPROACHES ALTERNATIVES

1. Classic (Frequentist) parametric estimation (OLS, MLE);
2. Bayesian parametric estimation (Metropolis, Metropolis-Hastings, Gibbs Sampler)
3. Neural Network modeling
4. Association rules (non-parametric)

ASSUMPTIONS CLASSIC METHODS (OLS, MLE)

Sampling from a population following a stable distribution with fixed but unknown parameters

- + Traditional and automated;
- Implies sampling from a distribution fixed, stable parameters
- Gives the same weight to older and newer observations in parameters' estimation (if no outliers)

Source: http://www.energy.alberta.ca/OilSands/pdfs/FS_SAGD.pdf

ASSUMPTIONS CLASSIC METHODS (OLS, MLE)

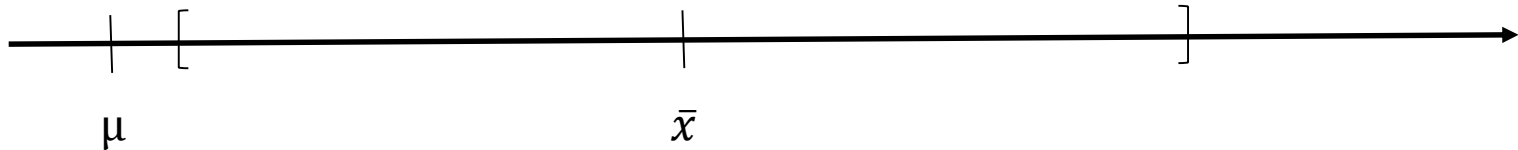
The model's parameters are fixed but unknown;

Implications:

- Older and newer observations have the same influence on the parameter estimates;
- No probability statements about the parameters because they are fixed:

95% confidence interval interpretation:

If 100 samples drawn from the same population, approximately 5 intervals will not contain the population mean.



ASSUMPTIONS BAYESIAN MCMC

The model's parameters are random variables and conditional on data:

$$Pr(H|E) = \frac{Pr(E|H) Pr(H)}{Pr(E)}$$

$$Pr(\theta|x) = \frac{f(x|\theta) \pi(x)}{\int f(x|\theta) \pi(x) dx} = \frac{(Sampling\ density\ of\ X\ given\ \theta)(Prior\ distribution\ for\ \theta)}{Marginal\ likelihood}$$

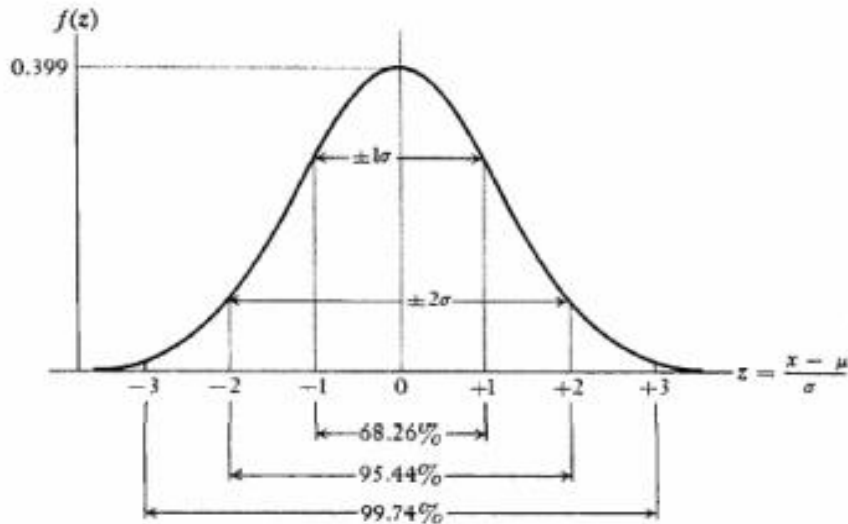
Implications:

- + **Newer observations can have higher influence on the parameters estimates that the older observations;**
- + Probability statement on the parameters that are conditional on observed data;
- + Bayesian simulation (MCMC) use random sampling to obtain approximate solutions when there is no closed-form solutions

ASSUMPTIONS BAYESIAN MCMC

95% credible interval:

There is a 95% chance that the parameter in the credible interval



BAYESIAN KEY PARAMETERS:

+ Prior distribution aggregates the information about the data parameters:

1. Informative prior = let your data drive ~100% estimation results:

- a) CME's survey results;
- b) Previous researches in the field;

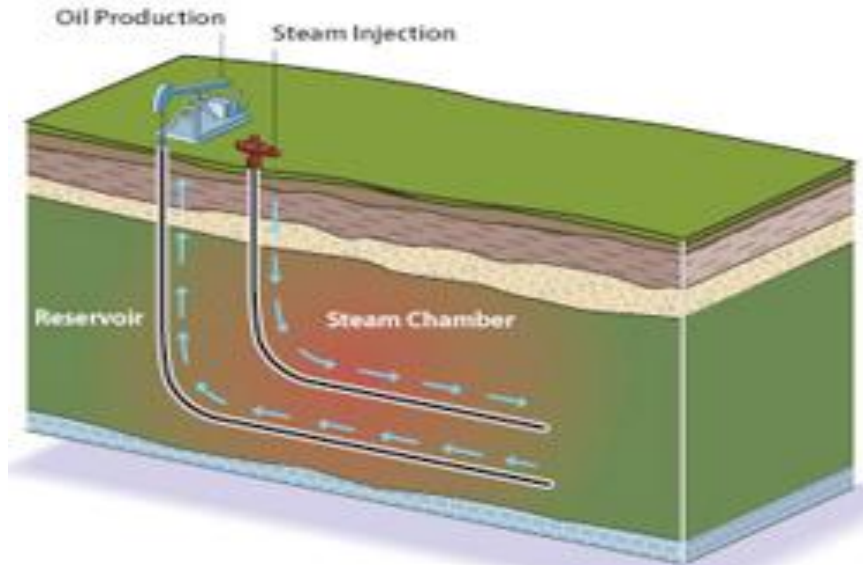
e.g., $\theta \sim N(\mu=1.5, \sigma=250)$

2. Non-informative, flat prior for no information available = update your prior for a new data:

e.g., $\theta \sim N(\mu=0, \sigma=10000000)$

APPLICATION MODELING SAGD EXAMPLE

Find the optimal values for the control variables that lead to maximum Oil Production.



Source: ©Canadian Centre for Energy Information 2009

→ Represent control variables that can be manipulated by operations:

1. Pump Speed
2. Short injection string
3. Long injection string
4. Casing gas
5. Header pressure
6. AR1 term

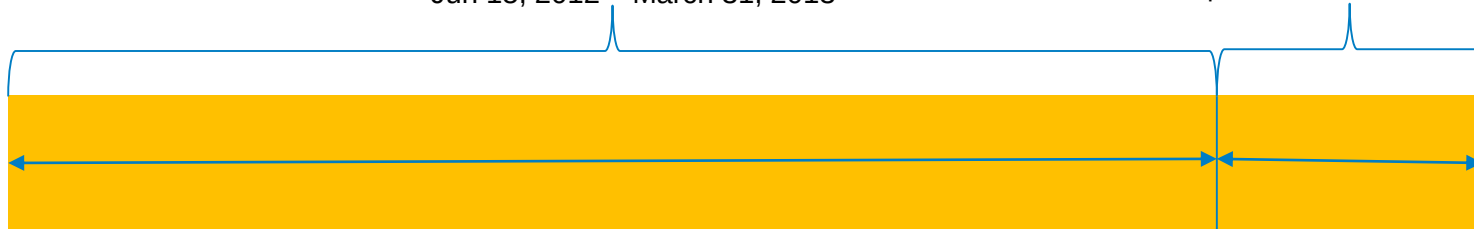
Dependent variables:
Oil (maximum)
Water (minimum)

DATA DIVISION:

Classic regression model:

Training Data :
Jun 15, 2012 – March 31, 2013

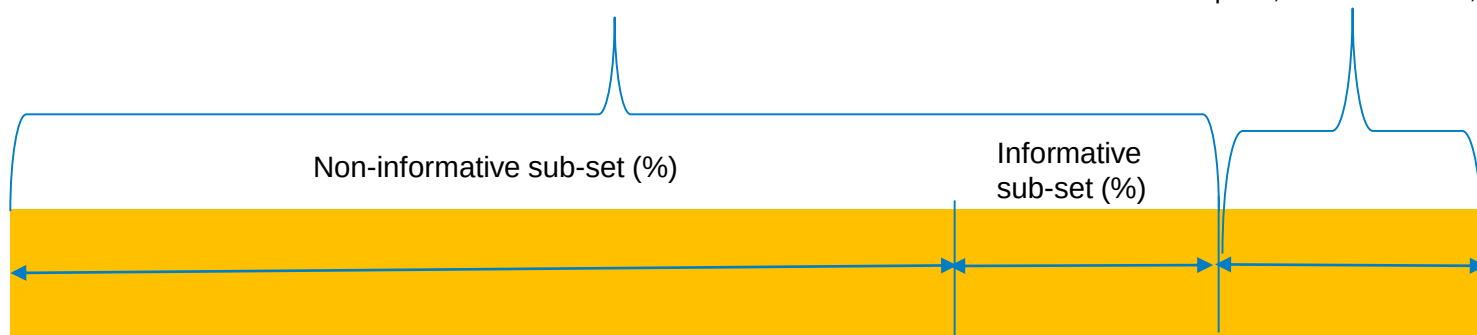
Validation Data:
April 1, 2013 – June 14, 2013



Bayesian regression model:

Training Data : Jun 15, 2012 – March 31, 2013

Validation Data:
April 1, 2013 – June 14, 2013



DATA MODEL

Stage 1: Non-informative priors and “old” data

$$Y_t | \beta_1, \dots, \beta_6, X_{1,t}, \dots, X_{5,t}, Y_{t-1} \sim N(\beta_1 X_{1,t} + \dots + \beta_5 X_{5,t} + \beta_6 Y_{t-1}, \sigma^2)$$

$$\beta_i \sim N(0, 10^6)$$

$$\sigma^2 \sim \text{Inv} - \text{Gamma}(\text{shape} = 2.001, \text{scale} = 1.001)$$

Stage 2: Using Stage 1 estimates as parameters for informative priors and “new” data

$$Y_t | \beta_1, \dots, \beta_6, X_{1,t}, \dots, X_{5,t}, Y_{t-1} \sim N(\beta_1 X_{1,t} + \dots + \beta_5 X_{5,t} + \beta_6 Y_{t-1}, \sigma^2)$$

$$\beta_i \sim N(\hat{\beta}_i^{\text{stage 1}}, \hat{\sigma}_i^{2 \text{ Stage 1}})$$

$$\sigma^2 \sim \text{Inv} - \text{Gamma}(\text{shape} = 2.001, \text{scale} = 1.001)$$

BAYESIAN KEY PARAMETERS

1. **NBI** = *Burn-in* = discarding initial observation in MCMC sample to minimize their effect on the posterior inference.

By default, NBI=1000;

2. **THIN** = Thinning = keeping every kth simulated draw from each sequence to reduce sample autocorrelation.

By default, THIN=1;

3. **NTU** = *Tuning* = number of iterations in the block at which MCMC evaluates the acceptance of the proposal

distribution. By default, NTU=500 and probability of acceptance ACCEPTTOL=0.075;

4. **PROPCOR** = controls options for constructing the initial proposal covariance matrix.

PROPCOR = QUANEW find numerically approximated covariance matrices at the optimum of the posterior density function with respect to all continuous parameters.

Stage 1: Non-informative priors and “old” data

```
proc mcmc DATA=SASDATA.TRAIN_DATA_90 diag=all dic
```

```
/* Key simulation parameters */
```

```
    propcov=quanew /* specifies a method for constructing an initial covariance matrix */  
    nbi=5000        /* the number of burn-in iterations to perform before saving parameter estimates chains */  
    ntu=5000        /* the number of iterations to use in each proposal tuning phase */  
    nmc=200000      /* the number of simulations */  
    thin=10         /* thinning rate of the simulation */  
    plots(smooth)=all seed=12345;
```

```
/* Model's parameters and their initial values */
```

```
parms (beta1 beta2 beta3 beta4 beta5 beta6) 0;  
parms sigma2 1;
```

```
/* Prior distribution assumptions */
```

```
prior beta1 beta2 beta3 beta4 beta5 beta6 ~ normal(mean=0, var=1e6);  
prior sigma2 ~ igamma(shape=2.001, scale=1.001);
```

```
/* Data model */
```

```
mu = beta1*X1 + beta2*X2 + beta3*X3 + beta4*X4 + beta5*X5 + beta6*Oil_Pair_1;  
model Oil_Pair ~ normal(mu, var=sigma2);  
run;
```

Stage 2: Using Stage 1 estimates as parameters for informative priors and “new” data

```
proc mcmc DATA=SASDATA.TRAIN_DATA_10 diag=all dic
/* Key simulation parameters */
    propcov=quanew
    nbi=5000
    ntu=5000
    nmc=200000
    thin=20
    plots(smooth)=all seed=12345;
/* Model's parameters and their initial values*/
parms (beta1 beta2 beta3 beta4 beta5 beta6) 0;
parms sigma2 1;
/* Prior distribution assumptions */
prior beta1 ~ normal(mean=-0.00327, var=0.000365);
prior beta2 ~ normal(mean=-0.00033, var=0.000417);
prior beta3 ~ normal(mean=0.2419, var=0.0190);
prior beta4 ~ normal(mean=0.000036, var=0.000029);
prior beta5 ~ normal(mean=-0.00004, var=0.000021);
prior beta6 ~ normal(mean=0.6340, var=0.0237);
prior sigma2 ~ igamma(shape=2.001, scale=1.001);
/* Data model */
mu = beta1*X1 + beta2*X2 + beta3*X3 + beta4*X4 + beta5*X5 + beta6*Oil_Pair_1;
model Oil_Pair ~ normal(mu, var=sigma2); run;
```

See details at:

[sashtml2.htm](#)

[HTML - Water Bayesian, Train, Valid, Similarity.html](#)

[TADPanel2.png](#)

[TADPanel5.png](#)

[TADPanel5.png](#)

[TADPanel8.png](#)

[TADPanel11.png](#)

[TADPanel14.png](#)

[TADPanel17.png](#)

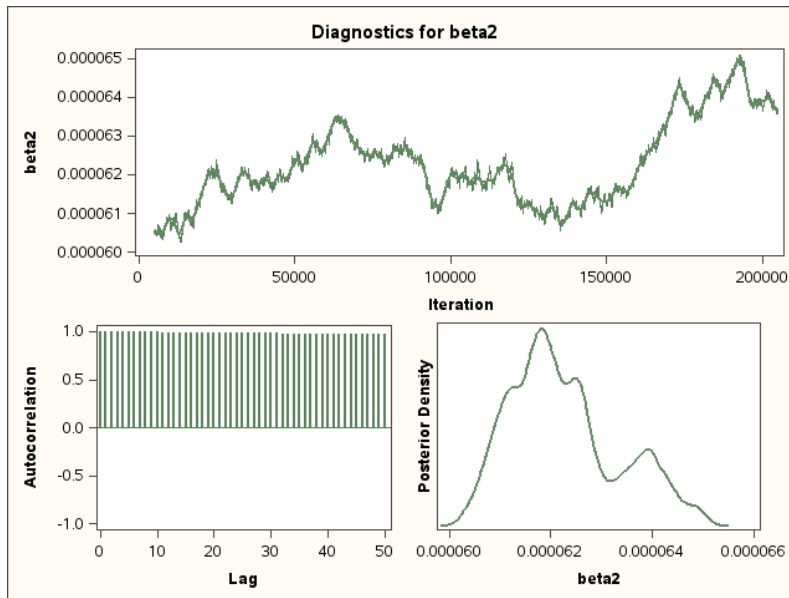
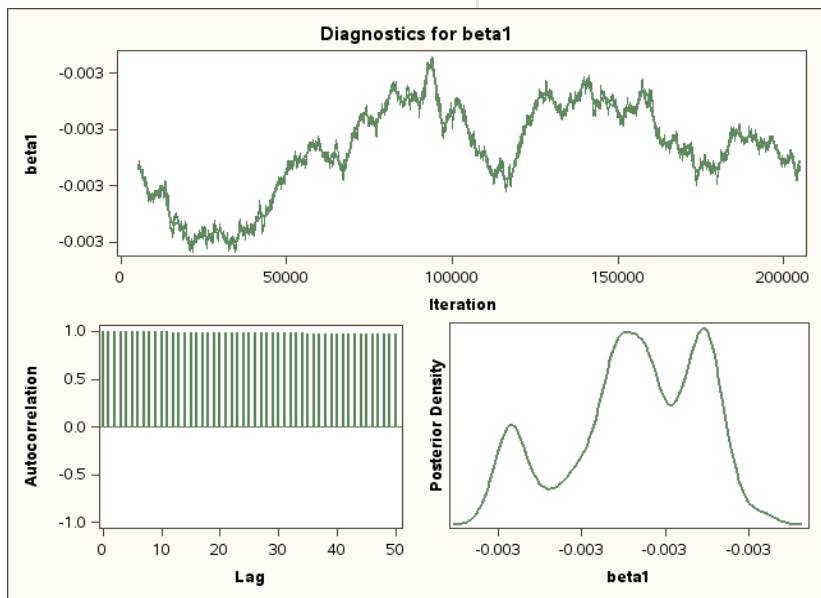
[TADPanel20.png](#)

BAYESIAN KEY DIAGNOSTICS: TRACE PLOT

Trace plot diagnostics:

1. Covariance stationarity: a) stable mean, b) constant variance;
2. Observations are spread out and do not stick to the same area
3. Low autocorrelation

BAYESIAN KEY DIAGNOSTICS: ISSUES TO WATCH



Problems

Multi-modal posterior distribution;
High auto-correlation in iterations

Potential Solutions

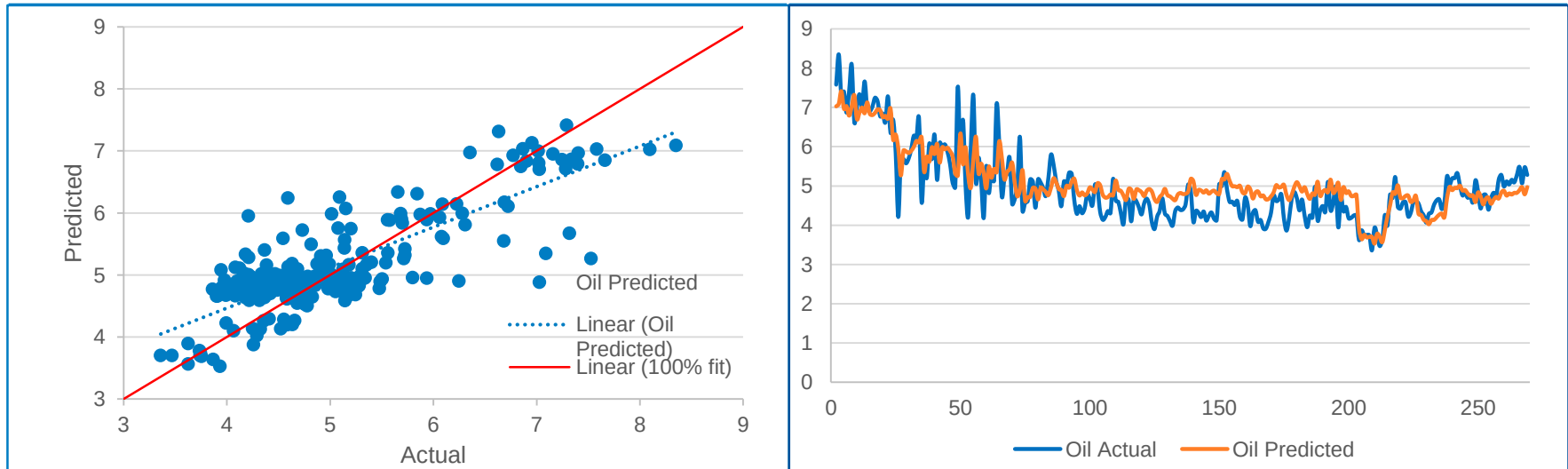
Increase thinning rate
Increase burn-in rate
Investigate data issues/increase a data sample

GOODNESS-OF-FIT

1. Deviance Information Criterion (DIC) to compare models developed for the same data set (AIC, BIC)
2. To compare models using different data sets

$$\text{Similarity Measure} = 1 - \frac{|\text{Actual Data} - \text{Predicted Data}|}{\text{Actual Data}}$$

3. Predicted Vs. Actual charts



Id	Model	Similarity
1.2	Frequentist MLE	<u>Similarity = 0.53 (validation)</u>
2.1	1 stage Bayesian: non-informative + 100% data	0.7186 (validation)
2.2	2 stage Bayesian: 1) non-informative priors+ 90% data; 2) 1 stage Informative priors + 10% of data	0.7536 (validation)
2.3	2 stage Bayesian: 1) non-informative + 75% data; 2) Informative + 25% of data	0.7144 (validation)
2.4	2 stage Bayesian: 1) non-informative + 60% data; 2) Informative + 40% of data	0.6845 (validation)
2.5	2 stage Bayesian: 1) non-informative + 50% data; 2) Informative + 50% of data	0.7349 (validation)

SIMILARITY MEASUREMENT

Model	Classic Regression Approach	Similarity Measure
1	Water_P1	93%
2	Oil_P1	53%
3	Water_P2	84%
4	Oil_P2	72%

Model	Bayesian Regression Approach	Similarity Measure
1	Water_P1	96%
2	Oil_P1	75%
3	Water_P2	93%
4	Oil_P2	91%

Maximum similarity measure is achieved when the training data split for the best fitting validation data set:

- a) 75-90% for Non-informative sub-set;
- b) 10-25% for Informative sub-set.

REFERENCES

Details: MCMC Procedure

http://support.sas.com/documentation/cdl/en/statug/63033/HTML/default/viewer.htm#statug_mcmc_sect018.htm